

Public Finance

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**Taxes on Risk Taking and Wealth
(Ch.23)**



Taxes on Risk Taking and Wealth

Capital income taxation affects:

- how individuals take risk
- how they realize and report capital gains
- how wealth is transferred
- how property is taxed



Motivation: Why Tax Capital Income?

Governments tax not only labor income, but also:

- returns on savings
- capital gains
- wealth transfers
- property values

These taxes shape decisions about saving, investing, taking risks, selling assets, and transferring wealth.

They also play a major role in distributional outcomes.



Introduction (1)

In this chapter, we focus on two aspects of taxation that might affect the savings decisions of taxpayers, other than government policies that directly alter the rate of return to savings.

The first is the taxation of risk taking. Individuals decide not only how much to save but also what form their savings will take.

- A particular tax policy that might affect risk taking is the taxation of capital gains, the earnings realized on the sale of capital assets.



Introduction (2)

Second, individuals can also be taxed not only on the return from their savings in each period but also on the amount of wealth they have accumulated through past savings.

- One form of wealth tax is transfer taxes, a set of taxes on gifts from one party to another, including the contentious estate tax.
- A second form of wealth tax is the property tax. Because the major source of savings for most people is their homes, property taxation can significantly affect the level of savings as well.



Basic Financial Investment Model

How do taxes on risk taking affect behavior?

Taxing gains and deducting losses increase risk taking because any tax on returns can be completely undone by taking more risk.

- Consider a \$100 investment opportunity that pays out \$120 half the time and \$80 half the time. This has an expected return of zero.
 - Expected return: The return to a successful investment times the odds of success, plus the return to an unsuccessful investment times the odds of failure.



Some Taxes on Risk Taking Can Be Undone

Table 23-1

Policy	Investment	Payoff if Win	Payoff if Lose	Tax Rate if Win	Tax Deduction if Lose	After-Tax Winnings	After-Tax Loss
(1) No tax	\$100	\$20	-\$20	0	0	\$20	-\$20
(2) Tax	\$100	\$20	-\$20	50%	50%	\$10	-\$10
(3) Loss offset	\$200	\$40	-\$40	50%	50%	\$20	-\$20

- In row 1, Betsey makes a \$100 investment that yields \$20 if the investment is successful and -\$20 if the investment fails.
- In row 2, the government imposes a 50% tax on her investment, so that Betsey now earns only \$10 if successful and loses \$10 if the investment fails.
- Betsey can undo the government's actions, as in row 3, by doubling her investment to \$200, leaving her with the same returns as in row 1, so that taxation has increased risk taking.



Real-World Complications: Less-Than-Full Tax Offset (1)

- Tax loss offset: The extent to which taxpayers can deduct net losses on investments from their taxable income.
- A taxpayer cannot simply undo government taxation by making increasingly risky investments because the losses from these investments will not be fully deductible from taxation.



Real-World Complications: Less-Than-Full Tax Offset (2)

Table 23-1

Policy	Investment	Payoff if Win	Payoff if Lose	Tax Rate if Win	Tax Deduction if Lose	After-Tax Winnings	After-Tax Loss
(1) No tax	\$100	\$20	-\$20	0	0	\$20	-\$20
(2) Tax	\$100	\$20	-\$20	50%	50%	\$10	-\$10
(3) Loss offset	\$200	\$40	-\$40	50%	50%	\$20	-\$20
(4) No loss offset	\$200	\$40	-\$40	50%	0	\$20	-\$40

- If the government does not allow individuals to deduct investment losses, as in row 4, then Betsey cannot undo government tax policy by investing more.
- Investing \$200 means that she has a 50% chance of earning \$40, which is \$20 after tax, and a 50% chance of losing \$40. Thus, she cannot offset the tax policy by raising her investment.



Real-World Complications: Less-Than-Full Tax Offset (3)

Table 23-1

Policy	Investment	Payoff if Win	Payoff if Lose	Tax Rate if Win	Tax Deduction if Lose	After-Tax Winnings	After-Tax Loss
(1) No tax	\$100	\$20	-\$20	0	0	\$20	-\$20
(2) Tax	\$100	\$20	-\$20	50%	50%	\$10	-\$10
(3) Loss offset	\$200	\$40	-\$40	50%	50%	\$20	-\$20
(4) No loss offset	\$200	\$40	-\$40	50%	0	\$20	-\$40
(5) Progressive tax	\$200	\$40	-\$40	75%	50%	\$15	-\$20

- Suppose that, as in row 5, the government imposes a 50% tax rate on income up to \$20 and 75% on income above \$20. The government allows any losses to be deducted from taxable income at the 50% rate.
- In this case, her potential winnings have a higher tax rate than her potential losses, so her potential winnings (\$15) are less than her potential loss (\$20).

Real-World: Less Than Full Loss Offset

Reality: losses are only partially deductible.

Consequences:

- Investor cannot simply scale up the gamble.
- Asymmetric taxation of gains and losses reduces risk-taking relative to the full-offset case.



Progressive Taxation and Risk Taking

Tax systems are typically progressive, with higher marginal tax rates as income rises.

With progressive income taxation:

- **Winning** investments may push an investor into a higher tax bracket → gains taxed at higher rates
- **Losing** investments reduce taxable income but often at lower marginal rates → losses deducted at lower rates

This asymmetry reduces the after-tax expected return to risky investment.



Evidence on Taxation and Risk Taking

- Due to complications such as progressive tax schemes and less-than-full tax offset, there is no clear prediction about how taxation will affect risk taking in the real world.
- What effect taxes have on risk taking is an empirical question.
- Unfortunately, there is very little evidence in this area.



Empirical Evidence on Risk Taking

Theoretical predictions are ambiguous due to:

- incomplete loss offset
- progressive tax systems
- treatment of realized vs unrealized losses

Evidence:

- Surprisingly little empirical work
- Existing studies suggest **modest** effects of taxation on risk-taking
- Effects depend heavily on details of tax code design



Taxation and Risk Taking: Labor Investment Applications

- Education is a risky investment.
- Each year of schooling increases earnings by about 7% on average, but the actual returns vary.
- Labor taxes affect the after-tax return to education.
 - Theoretically, higher taxation should increase investment in human capital, so that individuals can preserve the desired amount of net expected investment return.
- Progressive income taxes may discourage investment in education.
- Thus, the net impact of income taxation on human capital investments is ambiguous.



What Are Capital Gains?

Capital gain:

Difference between an asset's purchase price and its sale price.

Example:

Buy stock at €100

Sell at €140

→ Capital gain = €40

Key distinction:

Interest income is taxed on accrual

Capital gains are **taxed on realization** (when the asset is sold).



Accrual vs. Realization

Taxation on accrual:

Pay taxes each period on the return earned in that period.

→ clean, theoretically efficient, but harder to implement.

Taxation on realization (current system):

Pay taxes only when selling the asset.

→ creates timing incentives and strategic behavior.



Lock-In Effect

Capital gains taxes create an incentive to **delay selling** appreciated assets.

Why?

Tax is paid upon realization, so delaying payment increases the **present value** of after-tax returns.

Consequences:

- Reduced market liquidity
- Inefficient allocation (people hold assets too long)
- Tax preferences distort portfolio decisions



Arguments for Lower Capital Gains Taxes

1. Protects against inflation

- Nominal gains include inflation
- Without indexation, real gains are overstated

2. Improves efficiency of capital transactions

- Reduces lock-in
- Encourages rebalancing

3. Encourages entrepreneurship

- Many entrepreneurial returns appear as capital gains
- But Gruber notes: preferences are a *very blunt tool*



Arguments Against Preferences

1. Violates Haig–Simons

Equivalent economic income should be taxed the same.

2. Benefits are regressive

Capital gains overwhelmingly accrue to top income/wealth groups.

3. Encourages tax planning

Shifting income to capital gains

Using carried interest loopholes

Re-labeling labor income

4. Large revenue loss with limited new investment

Much of the benefit goes to past gains, not new activity.



Evidence on Taxation and Capital Gains 1

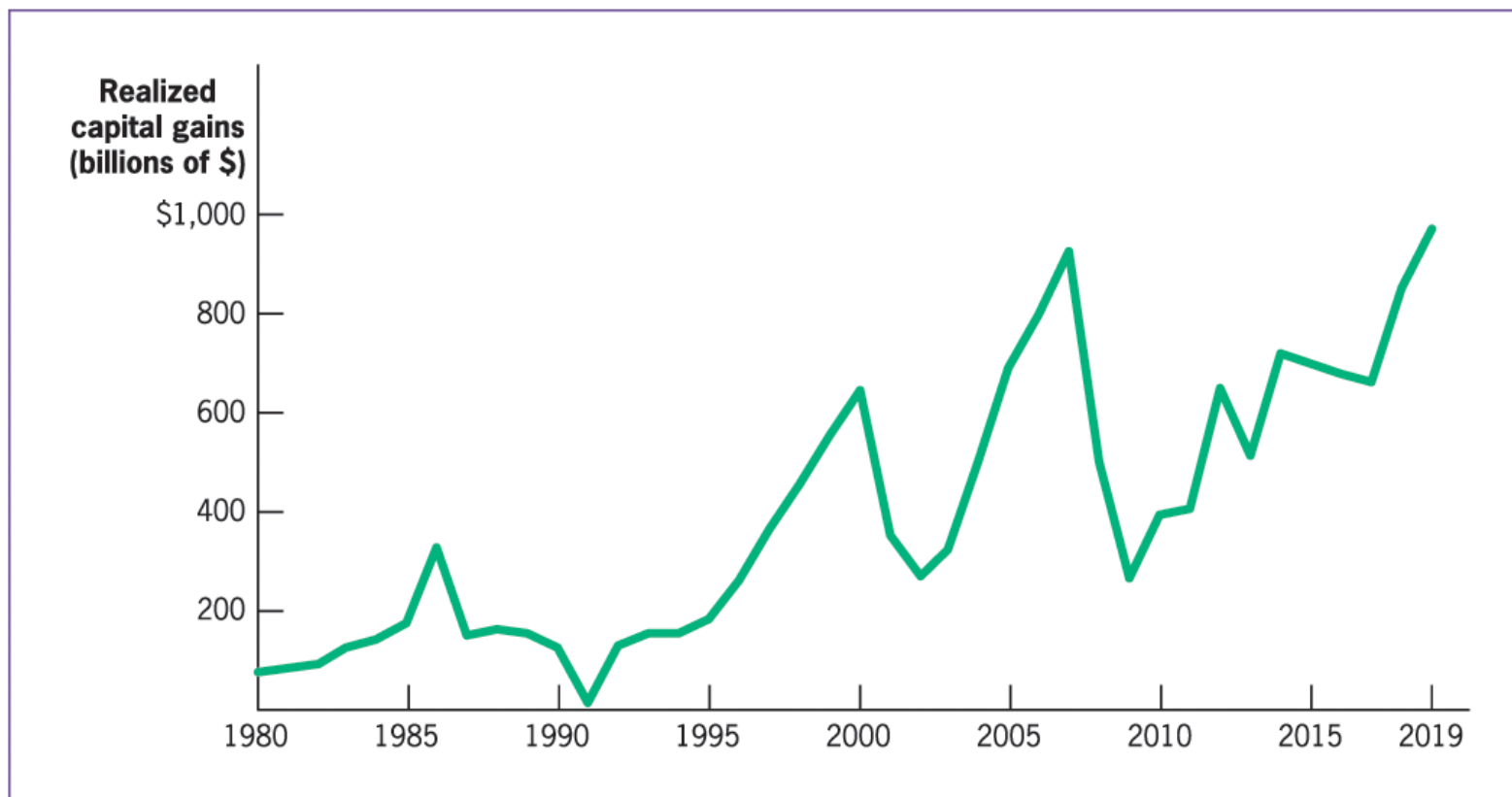
How does the capital gains tax rate affect capital gains realizations and thus capital gains tax revenue?

- Key question in determining how capital gains should be taxed.
- In 1986, a capital gains tax increase was announced, to go into effect in 1987 in the US.
- Capital gains realizations spiked in 1986 . . .
- but fell back to near their 1985 levels in 1987.
- The tax hike did not have a large long-run effect on tax revenues.



Evidence on Taxation and Capital Gains 2

Figure 23-1



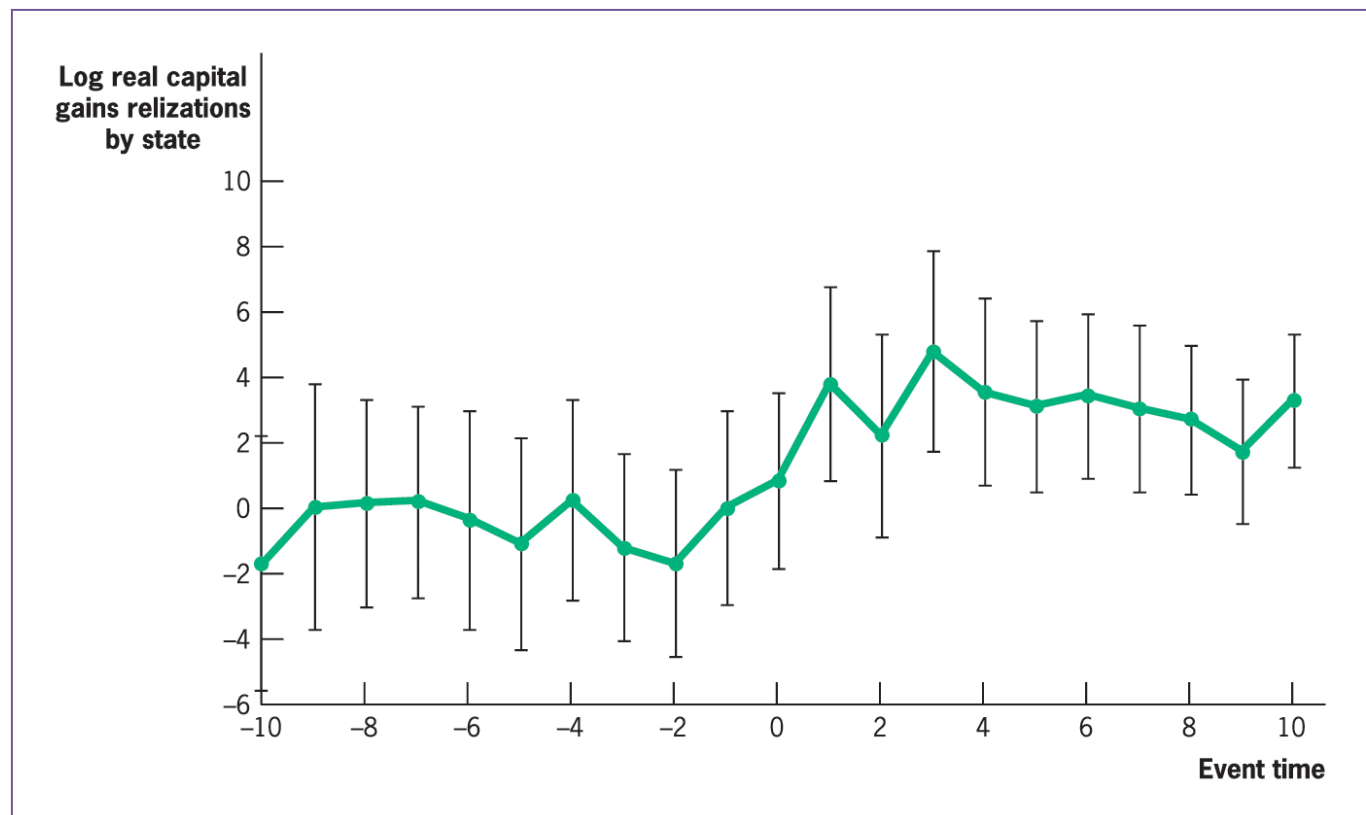
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There is a clear effect of tax changes on capital gains realizations. Realizations peaked in 1986 in anticipation of the increase in capital gains tax rates scheduled for 1987. Realizations peaked again during the stock market booms of the past decade.



Evidence on Taxation and Capital Gains 3

Figure 23-1



The figure (Agersnap and Zidar, 2020) shows the relationship between the “event time,” which is set at zero at the point at which states change their capital gains taxation, and the “net of tax” rate (one minus the tax rate). The figure demonstrates that capital gains realizations respond to tax changes. However, their calculations suggest a long run elasticity of capital gains with respect to the tax rate of -0.3 to -0.5 , fairly inelastic. The authors conclude that increasing the capital gains tax rate would lead to net increases in revenue.



Transfer Taxation

Transfer taxes are an important type of capital tax.

- Transfer tax: A tax levied on the transfer of assets from one individual to another.
- Gift tax: A tax levied on assets that one individual gives to another in the form of a gift.
- Estate tax: A tax levied on the assets of the deceased that are bequeathed to others.



Transfer and Wealth Taxes (% of Government Revenue)

The use of transfer taxes and wealth taxes varies widely around the world.

Table 23-2

	Transfer Taxes	Wealth Taxes	Transfer and Wealth Taxes
Australia	0.0%	0.0%	0.0%
Canada	0.0	0.1	0.1
Finland	0.7	0.0	0.7
France	1.4	0.2	1.6
Germany	0.5	0.0	0.5
Japan	1.3	0.0	1.3
Norway	0.0	1.5	1.5
Spain	0.6	0.5	1.1
Switzerland	0.6	4.7	5.3
United Kingdom	0.7	0.0	0.7
United States	0.5	0.0	0.5
OECD average	²⁵ 0.3	0.5	0.8



Why Tax Wealth? Arguments for the Estate Tax

Why tax wealth rather than income?

- Extremely progressive means of raising revenue.
- Necessary to avoid the excessive concentration of wealth and power in a few wealthy dynasties.
- Allowing children of wealthy families to inherit all their parents' wealth saps them of all motivation to work hard and achieve their own success.



Arguments Against the Estate Tax 1

Why not tax wealth?

- A “death tax” is cruel: It is morally inappropriate to tax individuals upon their death.
 - However, death is neither a necessary nor a sufficient condition to trigger transfer taxes:
 - Transfers between living persons can trigger gift taxes.
 - 98% of those who die pay no estate tax.
- The estate tax amounts to double taxation: You are taxed on income when you earn it, and then your children are taxed on it again when you die.
 - This is common; sales tax also amounts to double taxation.
 - Does not necessarily reduce savings.



Arguments Against the Estate Tax 2

- Administrative difficulties: To afford the tax, you may be forced to sell the asset.
 - While this is an issue in theory, it is not clear that it is a major issue in practice.
- Compliance and fairness: Only those too unsophisticated to avoid the tax end up paying it.
 - There are enough loopholes, such as trust funds, that some refer to the estate tax as a “voluntary tax.”



Example: Who Pays the Estate Tax?

- ~top of the wealth distribution
- ~98% of estates owe no estate tax.
- Liability is concentrated in the top of the wealth distribution.

This explains why the estate tax is politically salient despite affecting few taxpayers.

Takeaway:

Estate tax is **highly progressive**, but politically unpopular.



Property Taxation

- Property taxation is the major source of revenue for local governments.
 - Property tax: A tax levied on the value of real estate, including the value of land and any structures built on it.
- Tax burden based on assessed property value, but homeowners pay taxes only on the assessment ratio.
- Assessment ratios vary widely across the world.



Types of Property Taxation

Two important distinctions in levying property taxes:

1. Residential homes versus businesses

- Some argue that to encourage economic development, business property taxes should be lower than residential property taxes.

2. Land versus improvements

- Land is in inelastic supply, so it should be taxed.
- Hard to separate land value from structure value, so structures and land are both taxed in practice.



Property Tax Breaks: Examples

States frequently offer preferential tax treatment to businesses.

Examples from Gruber:

- Washington offered Boeing an \$8.7B tax break
- New York offered Alcoa \$5.6B
- Wisconsin subsidized Foxconn heavily

Local benefit: higher employment and property values

National outcome: zero net gain — just **state-level competition**



Wealth Concentration: Long-Run Trends (Saez & Zucman)

Modern research documents a strong rise in wealth concentration:

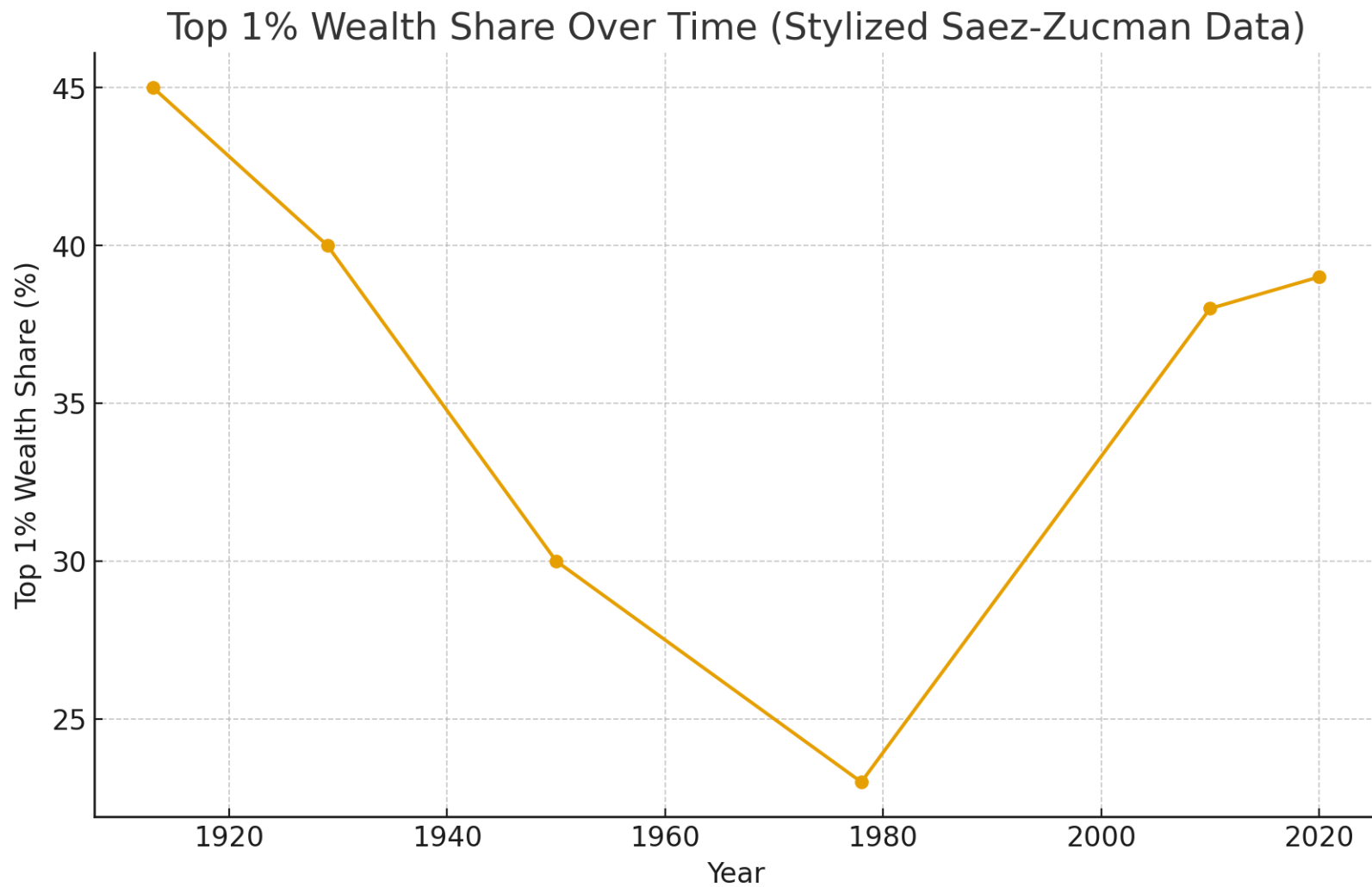
- Top 1% wealth share has increased significantly since the 1980s
- Top 0.1% share has risen even faster
- Driven by capital income accruing disproportionately to high-wealth households
- Wealth grows faster than income for rich households ($r > g$)

Implication:

Capital taxation debates today are inseparable from the distribution of wealth.

Source: Saez & Zucman (2019); Saez & Zucman (2016); WID.world (2022); OECD (2018).





Source: Saez & Zucman (2019); Saez & Zucman (2016); WID.world (2022); OECD (2018).

How Countries Tax Wealth

Cross-country variation:

- **Net wealth taxes** still exist in a few countries (e.g., Spain, Norway). Several major European countries abolished them (France, Denmark, Sweden) due to administrative and avoidance concerns
- **Estate/inheritance taxes** remain common but with very different rates and exemptions
- Wealth taxation often complements capital income taxation, not replaces it

Key point:

The world has shifted *away* from annual wealth taxes and *toward* inheritance taxation + capital gains taxation.

Source: Saez & Zucman (2019); Saez & Zucman (2016); WID.world (2022); OECD (2018).



Efficiency vs Equity in Wealth Taxation

Saez's framework highlights two central trade-offs:

Equity:

- Wealth is more concentrated than income
- Capital income grows faster for high-wealth households
- Untaxed wealth compounds over generations → dynastic inequality

Efficiency:

- Harder to measure/track wealth than income
- Risk of avoidance, offshoring, valuation disputes
- Annual wealth taxes risk taxing illiquid households

Policy conclusion: Wealth taxation is feasible, but administration and valuation are the main bottlenecks.

Source: Saez & Zucman (2019); Saez & Zucman (2016); WID.world (2022); OECD (2018).



Why This Matters for Capital Income Taxation

This chapter ties directly into the global debate:

- Taxing **capital gains** and **inheritance** is a practical way to tax wealth
- Preferential capital gains treatment amplifies wealth concentration
- Step-up in basis at death = large implicit tax break for the wealthy
- Transfer taxes are progressive but politically unpopular
- Many countries rely more on **capital income taxation** than wealth taxes

Big picture:

Capital taxation shapes long-run inequality more than labor taxation.

Source: Saez & Zucman (2019); Saez & Zucman (2016); WID.world (2022); OECD (2018).



Capital Income Taxation in Italy

Italy taxes most capital income at **proportional rates**, separate from the IRPEF schedule.

Main components:

- **Financial income** (interests, dividends, capital gains): **26%** standard rate
- **Government bonds (BTP, BOT): 12.5%** preferential rate
- **Capital gains on equities:** Generally **26%**, taxed on realization

Investment funds: Taxed at **26%**, with special rules for funds holding government securities

Key point: Italy's capital income taxation is *simpler* and more *flat* than labor taxation.

Italy: Inheritance and Gift Tax

Italy has **one of the lowest inheritance taxes in Europe.**

International comparison:

- Italy collects **very little** from inheritance taxation
- Rates and taxable bases are much lower than France, Belgium, Germany, Spain
- In practice, many estates pay **zero** inheritance tax

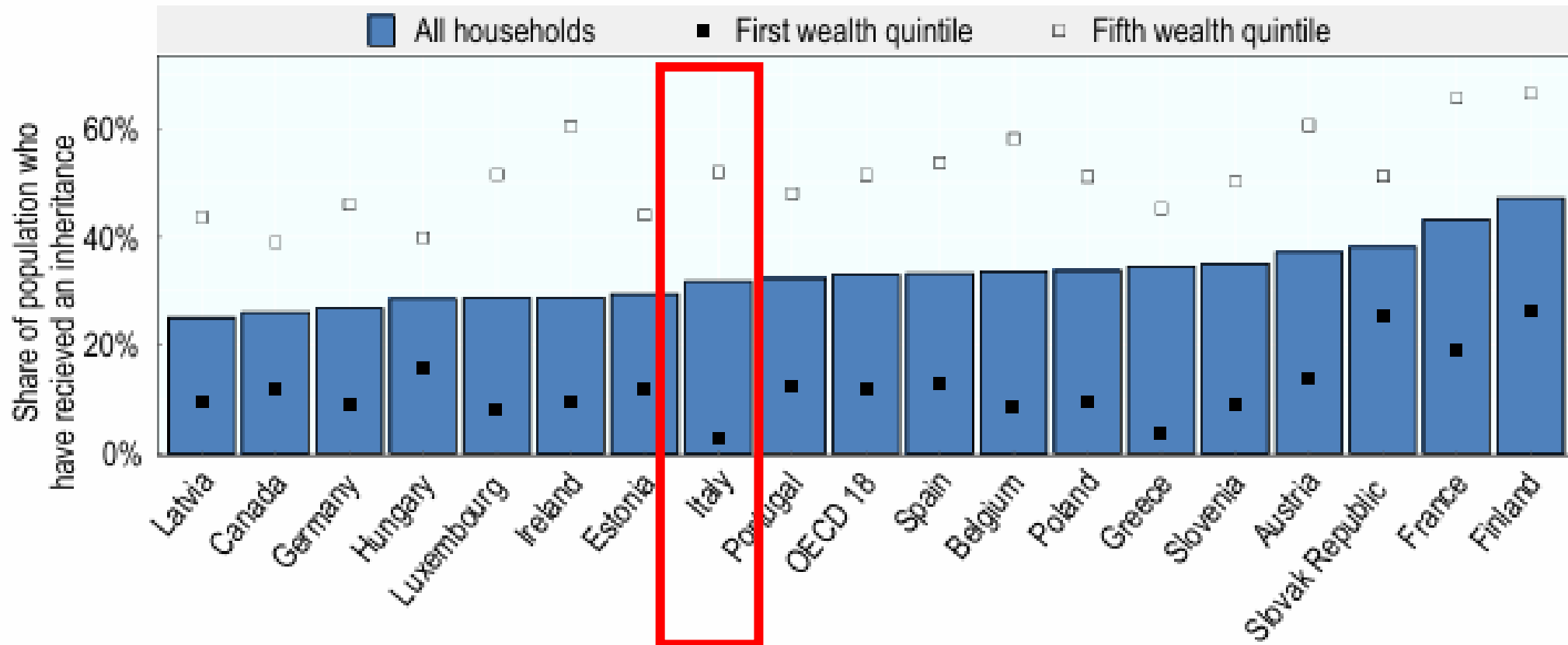
Implication:

Italy relies far less on transfer taxation than most advanced economies.

Source: OECD Revenue Statistics; OECD Tax Policy Database; Italian Ministry of Economy & Finance (MEF)



Share of population that has received an inheritance or a substantial gift, 2015



Source: OECD Wealth distribution database, 2018



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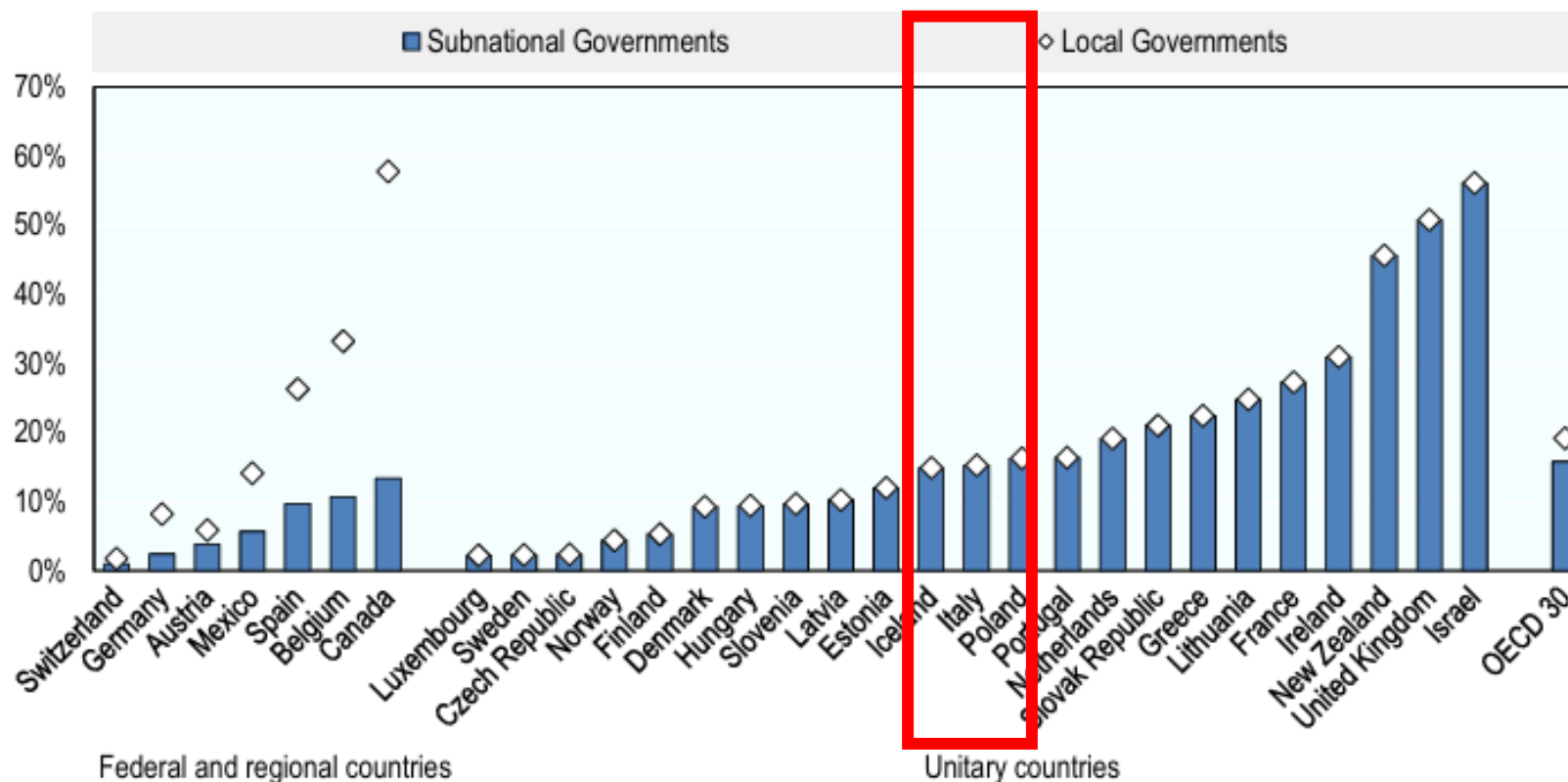
Italy: Property Taxation (IMU)

- Capital income tax rates vary widely across OECD countries.
- Countries with generous mandatory retirement systems show low sensitivity of saving to tax incentives.
- Capital shares and wealth concentration have increased globally.
- Despite different tax systems, household saving rates are relatively stable across countries.
- In Italy, IMU is not paid on the primary residence, reflecting distributional and political considerations, widespread homeownership, and liquidity concerns for households whose wealth is mainly housing.
- Overall, behavioral and institutional factors appear to matter more than tax incentives for saving decisions.

Source: OECD Revenue Statistics; OECD Tax Policy Database; Italian Ministry of Economy & Finance (MEF)



Recurrent taxes on immovable property as a % of local and state government's revenues, 2019



Note: Local and state revenues are consolidated to reflect own-source revenue (defined as total revenue minus the inter-governmental transfer revenue of that government level). The allocation of revenues between different levels of government is specified in the Fiscal Decentralisation Database. Values as of 2019 or closest year with available data.

Exercises



MC 1

With a proportional tax on risky returns and full loss offset, optimal risk-taking will:

- a) Decrease because expected returns fall
- b) Remain unchanged relative to no taxation
- c) Increase because the government bears all risk
- d) Increase only if the tax rate is low



MC 2

If losses are only partially deductible, taxing risky returns will typically:

- a) Increase risk-taking
- b) Leave risk-taking unchanged
- c) Reduce risk-taking
- d) Eliminate investment risk



MC 3

What is the main efficiency cost of realization-based capital gains taxation?

- a) Lower tax revenue
- b) Higher administrative costs
- c) The lock-in effect
- d) Reduced progressivity



Realization-based capital gains taxation creates a lock-in effect because:

- a) Investors face higher statutory tax rates
- b) Taxes are paid only when assets are sold
- c) Losses cannot be deducted
- d) Capital gains are taxed progressively



MC 5

The economic incidence of estate taxes typically falls on:

- a) Middle-income households
- b) Consumers through higher prices
- c) Heirs of wealthy individuals
- d) Workers through lower wages



True/False (please provide a justification)

- 1) With full loss offset taxation of risky assets necessarily discourages risk-taking.
- 2) Realization-based capital gains taxation encourages investors to hold appreciated assets longer than is socially optimal.
- 3) Estate taxes are typically justified by both equity and efficiency considerations.



Exercises

EXERCISE 1

An individual can invest €1 in a risky asset. The asset yields:
a return of +50% with probability $\frac{1}{2}$
a return of -50% with probability $\frac{1}{2}$

The government taxes capital income at rate of 30%.

- a) Compute the investor's expected return before tax.
- b) Compute the expected return after tax under:
 - full loss offset
 - no loss offset
- c) In which case is risk-taking more strongly discouraged? Explain intuitively.
- d) Explain how this example illustrates the efficiency effects of taxing risky returns



EXERCISE 2

Consider an investor who owns an asset that has appreciated in value. Capital gains are taxed at rate τ only upon realization.

- a) Explain why realization-based capital gains taxation creates an incentive to delay selling appreciated assets.
- b) Compare this system with accrual taxation in terms of efficiency.
- c) Discuss one administrative reason why governments often prefer realization-based taxation despite its inefficiency.
- d) Briefly explain how the lock-in effect can affect aggregate economic efficiency.

